

E-COMMERCE • BUSINESS BRIEFING

# Product Recommendation Systems in E-Commerce

*From Browsing to Buying — The Business Case*

A Briefing for Business Stakeholders

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# Helping shoppers find what they didn't know to search for



## The discovery problem

Catalogs hold millions of items; each shopper realistically sees only a handful.



## Choice overload

Too many options causes hesitation — and hesitation costs sales.



## The business case

Better matching lifts conversion, raises average order value, and deepens retention.

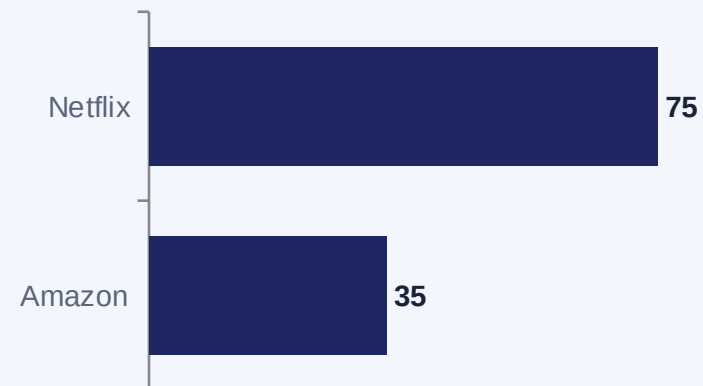
**~35%**

of Amazon sales linked to  
recommendations

**~75%**

of Netflix viewing driven by  
recommendations

### Activity driven by recommendations



*Widely cited industry estimates — directional, not exact.*

# Predict relevance, then rank



A recommender predicts which products a customer is most likely to want next — and presents them in the right order, at the right moment.



## Input

Who the customer is, what they've browsed and bought, and the current context.



## Model

Scores how relevant each product is to this customer right now.



## Ranked output

A short, ordered list of the best products to show.

*The goal is relevance: the right products, in the right order, surfaced automatically.*

# The same engine, many storefront surfaces



## Personalized

"Recommended for you," tuned to each individual.



## Frequently bought together

Complementary add-ons at the cart and product page.



## Trending & popular

What's hot right now — great for new or undecided shoppers.



## Similar items

"More like this" to keep the browsing journey going.



## Recently viewed

Easy re-entry to products a shopper already considered.

## CORE APPROACHES

# How a recommender decides what to show

*Five families of techniques — each suited to different goals and data.*



### Content-based

Match items by their attributes (brand, category, price).



### Collaborative filtering

“Customers like you also liked...” — learns from behavior.



### Embeddings

Compact patterns of taste that power fast, accurate matching.



### Hybrid

Blend several methods for the best of each.



Modern systems increasingly add deep learning and session-aware models to read intent in the moment.

## Two kinds of signal: what users say vs. what they do



### Explicit signals

*Users tell us directly*

- ✓ Star ratings
- ✓ Written reviews
- ✓ Likes & wishlists
- ✓ Survey preferences



### Implicit signals

*We observe behavior*

- ✓ Clicks & taps
- ✓ Purchases
- ✓ Time spent viewing
- ✓ Add-to-cart & removals



Most signal is implicit — abundant, but quieter. The richest systems combine both.

## THE COLD-START PROBLEM

# When there's no history to learn from



### New user

New customers arrive with no history — we don't yet know their taste.



### New item

New products have no sales or views — nothing to learn from yet.

## Common workarounds



Lead with popular & trending items



Ask a few onboarding preferences



Use product attributes for brand-new items



Lean on context: location, device, time

## Test offline to shortlist, test online to decide



### Before launch (offline)

- Does the model rank good items near the top?
- Tested on past data, fast and cheap



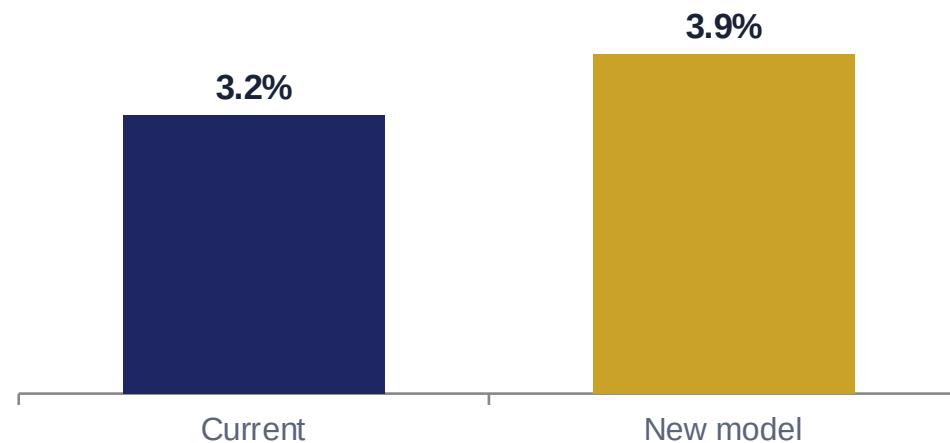
### In production (online)

- Click-through, conversion, revenue, AOV
- Proven with controlled A/B tests



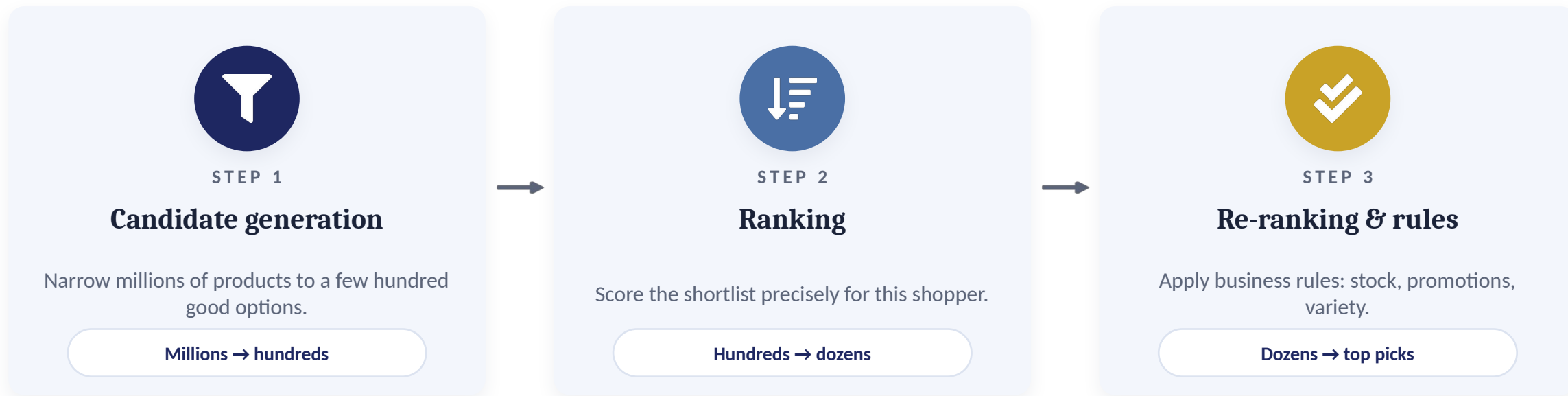
Offline scores guide the shortlist; live A/B tests decide what ships.

A/B test — conversion rate (%)





# Retrieve broadly, rank precisely, then apply the rules



Heavy preparation runs in batch overnight; the final pick happens in real time as the page loads.

# Relevance is not the only thing that matters



## Filter bubbles

Showing only the familiar can narrow what customers discover.



## Popularity bias

Best-sellers crowd out promising niche products.



## Diversity vs. relevance

Too safe feels repetitive; too varied feels random.



## Privacy

Personalization must respect consent and data limits.



## Fairness

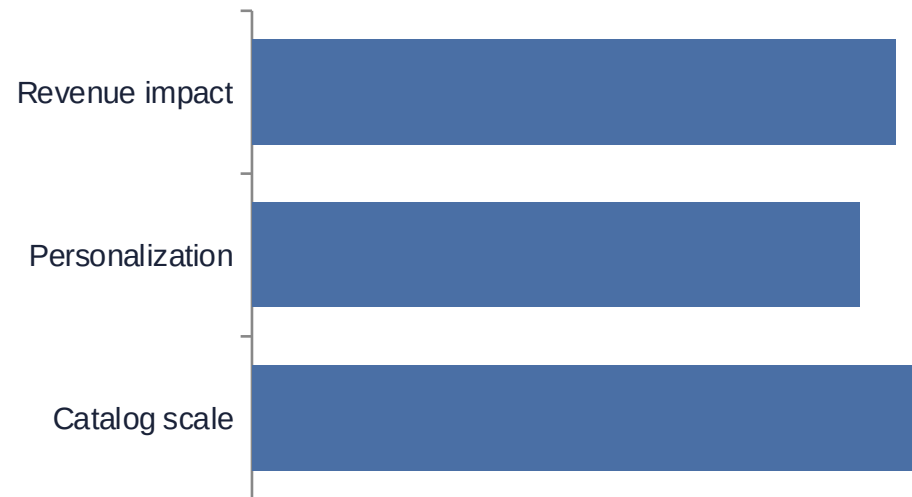
Sellers and customer groups deserve even-handed exposure.

# Amazon — item-to-item recommendations

## The approach

- 1 Pre-compute which products are bought or viewed together
- 2 At page load, instantly surface items similar to what you're seeing
- 3 Scales to enormous catalogs and powers “bought this also bought”

## Illustrative strength profile



*A simple, scalable idea that became a benchmark for retail recommendations worldwide.*

## KEY TAKEAWAYS

# What to carry out of this session

- 1 Recommendations turn a vast catalog into a personal storefront
- 2 They move the metrics leaders care about: conversion, AOV, retention
- 3 Start simple; sophistication should follow proven value
- 4 Always validate with live A/B tests before scaling
- 5 Balance relevance with trust: diversity, privacy, fairness

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***Let's discuss where recommendations can move your numbers.***

Thank you